

Alok Silswal

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Education

Cluster Innovation Centre, University of Delhi

August 2024 – Present

BTech in Information Technology and Mathematical Innovation

- GPA: 9.27 / 10.0
- **Coursework:** Data Structure & Algorithm, Analysis & Design of Algorithms, Probability and Statistics, Operating System, Computer Architecture, Linear Algebra, Discrete Mathematics, Calculus

SKILLS SUMMARY

Languages: C++, Python, Kotlin, HTML, CSS, JavaScript

Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib

Tools: Canva, Word, Excel, PowerPoint

Platforms: Visual Studio Code, Google Colab, IntelliJ IDEA, Google Earth Engine

Projects

Species Distribution Modeling (SDM) – *Gyps indicus*



- Built an end-to-end presence-only SDM using Google Earth Engine (GEE) and machine learning (Kernel Density Estimation) for habitat suitability mapping.
- Sampled 11K+ occurrence records and generated 77K background points in GEE; extracted climatic (WorldClim) and elevation (SRTM) variables at ~1 km scale.
- Applied feature preprocessing (scaling, transformation) and evaluated model using AUC = 0.8664 (presence vs background).
- Generated Gaussian-smoothed occurrence density heatmaps and predicted habitat suitability maps, visualized in Python and GEE for spatial validation.

Tech Stack: Python, JavaScript (Google Earth Engine), scikit-learn, Pandas, NumPy, Matplotlib

JumpYT – YouTube Timestamp Bookmarking Browser Extension



- Built a browser extension integrated into YouTube's UI for saving, labeling, editing, and navigating timestamps.
- Implemented dynamic DOM injection and event-driven logic while preserving native player behaviour.
- Engineered per-video persistent storage using browser local storage with session consistency.
- Validated extension permissions, manifest configuration, and correctness using automated linting.

Tech Stack: JavaScript (ES6), HTML, CSS, Browser Extension APIs

Experience

• **Vice-Chancellor Internship Scheme**

Nov 2025 – May 2026

Department of Statistics, University of Delhi

Redesigned, refactored, and maintained the **frontend** of the departmental website, improving reusability, scalability, responsiveness, and simplifying updates for non-technical maintainers.

Tech Stack: HTML, CSS, JavaScript, PHP

• **Open Source Contributor — AnkiDroid**

2025 – Present

Contributed bug fixes and feature improvements to AnkiDroid through multiple merged GitHub pull requests.

Tech Stack: Kotlin, Android, Git, GitHub

Achievements

- Secured **3rd place** in Hack4VikshitBharat Hackathon
- Secured **2nd place** at UDGAAR Inter-College Anti-Addiction Quiz Competition.